

WEEK-1

OUT-PUT EXERCISE-5

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:E:\IntelliJ2026\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=5530
Employee Added
Employee Added
Employee Added

Employees:
101 Rahul Manager 70000.0
102 Priya Developer 60000.0
103 Kiran Tester 50000.0

Search Employee:
102 Priya Developer 60000.0
Employee Deleted

After Deletion:
101 Rahul Manager 70000.0
103 Kiran Tester 50000.0

Process finished with exit code 0
```

Explanation:

Time Complexity

Operation	Time Complexity
Add Task	$O(n)$
Search Task	$O(n)$
Traverse Tasks	$O(n)$
Delete Task	$O(n)$

Advantages of Linked Lists over Arrays

Linked Lists	Arrays
Dynamic size	Fixed size
Efficient insertion/deletion	Shifting elements required
No contiguous memory needed	Requires contiguous memory
Easy expansion	Difficult to resize